

Technical Report

English Proficiency Across Europe and Italy Datasets, Preprocessing, and Methodological Considerations

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1 Introduction

This technical report documents the datasets, preprocessing procedures, and methodological considerations underlying the development of the interactive data visualization project initially introduced in the project proposal.

The project explores English proficiency patterns across Europe and Italy through the integration of multiple heterogeneous public datasets related to English proficiency, educational exposure, regional socio-economic indicators, and educational quality metrics. The primary objective of the visualization system is exploratory: to identify spatial disparities, temporal trends, and potential associations between English proficiency outcomes and selected contextual variables.

While the initial proposal defined the general analytical framework and the intended visualization objectives, the implementation process required several methodological adaptations related to data availability, harmonisation constraints, missing observations, and differences in temporal and spatial granularity across datasets. Consequently, certain aspects of the final visualization system differ from the original proposal, particularly regarding variable selection, preprocessing procedures, and visualization scope.

The final project was developed as a web-based interactive dashboard combining choropleth maps, comparative visualizations, temporal trend analysis, and exploratory filtering tools. The dashboard additionally includes interactive year selection, comparative country rankings, European aggregate indicators, and dedicated comparisons between Italy and broader European trends. Particular attention was given to dataset consistency, temporal alignment strategies, variable standardisation, and the treatment of incomplete observations.

Because the project combines datasets originating from different institutions, years, and reporting methodologies, substantial preprocessing and harmonisation operations were required before visualization development. The purpose of this technical report is therefore to document the structure and quality of the datasets used, the preprocessing operations performed, and the methodological limitations that may affect interpretation of the visualized results.

Each section of the report corresponds to one component of the final visualization project and presents the associated datasets, transformations, descriptive insights, and limitations.

2 Part 1 — ESL Exposure and English Proficiency Across Europe

2.1 Dataset Origin, Structure, Variables, and Metadata

This section investigates the relationship between exposure to English as a second language (ESL) in upper-secondary education and later English proficiency outcomes across European countries.

Two primary datasets were integrated for this sub-project.

The first dataset is the *EF English Proficiency Index* (EF EPI), published annually by Education First. The dataset contains country-level English proficiency scores derived from voluntary online English tests completed by adult participants. The raw dataset initially contained 246 observations across 38 countries between 2017 and 2024. Main variables include country name, EF EPI score, ranking, and year. No missing values were identified in the retained EF EPI observations.

The second dataset originates from Eurostat: *“Pupils by education level and modern foreign language studied”*. This dataset contains yearly statistics describing the proportion and number of pupils studying foreign languages across European educational systems. The raw Eurostat dataset contained 12 133 observations and 18 variables covering 41 countries between 2012 and 2024. Relevant variables included country identifier, year, educational level, language studied, and percentage of pupils learning English.

Because Eurostat datasets were initially structured in wide format, preprocessing included reshaping operations transforming yearly columns into long-format country-year observations suitable for temporal analysis and dataset merging. Additional filtering operations retained only observations corresponding to English language learning indicators (`language == ENG`), percentage-based measurements (`unit == PC`), and total population aggregates in order to ensure cross-country comparability.

Several cleaning and harmonisation procedures were additionally implemented. Country names and identifiers were standardised to ensure consistency across sources. Specific harmonisation rules were introduced for naming inconsistencies such as “Czech Republic” to “Czechia”, “Türkiye” to “Turkey”, and “UK” to “United Kingdom”. ISO-3 country codes were subsequently generated and used as the primary integration key across datasets and dashboard visualisations.

The preprocessing pipeline also removed incomplete observations, invalid country identifiers, and irrelevant language categories. After cleaning, the Eurostat dataset retained 365 country-year observations across 34 countries, while the cleaned EF EPI dataset retained 238 observations across 36 countries.

To account for delayed educational effects, a temporal lag alignment strategy was implemented. ESL exposure measured at year t was associated with EF EPI proficiency outcomes measured at year $t + 4$. This lag structure was introduced to approximate the delayed impact of secondary-school English exposure on later adult proficiency outcomes.

Additionally, percentile standardisation was applied yearly in order to improve comparability between variables measured on different scales. Country-level percentile rankings were computed separately for ESL exposure and EF EPI proficiency scores.

A derived variable, `gap_pct`, was constructed as:

$$\text{gap_pct} = \text{EF percentile} - \text{ESL exposure percentile}$$

Positive values indicate countries performing above expected educational exposure levels, whereas negative values indicate relative underperformance.

After preprocessing, harmonisation, percentile computation, and temporal lag alignment, the final analytical dataset contained 365 country-year observations and 7 variables. However, only 165 observations corresponded to complete matched country-year pairs across both datasets, corresponding to an effective EF matching rate of approximately 45%.

The relatively low matching rate mainly results from temporal lag alignment constraints, heterogeneous temporal coverage across datasets, and missing country-year observations in the

original sources.

Dataset	Years	Countries	Observations	Main Variables
EF EPI	2017–2024	38	246	Score, ranking
Eurostat ESL	2012–2024	41	12 133	ESL exposure indicators

Table 1: Main datasets integrated for Part 1 of the project.

Figure 1 illustrates the spatial distribution of the derived `gap_pct` variable across European countries for the latest available year after preprocessing and dataset harmonisation. Certain countries are absent due to missing or incomplete country-year observations removed during preprocessing and temporal lag alignment.

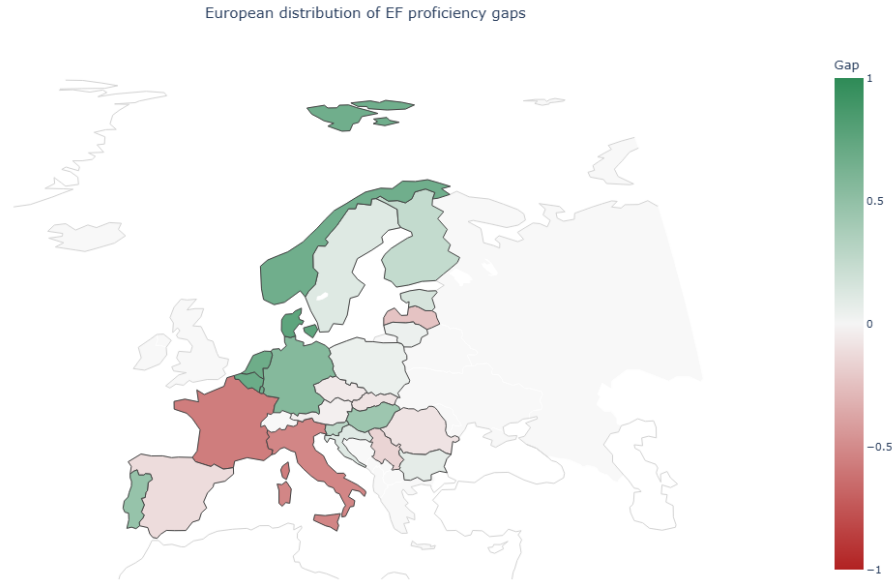


Figure 1: Percentile-based proficiency gap across European countries for the latest available year after preprocessing and temporal lag alignment. Positive values indicate countries whose EF EPI percentile exceeds their ESL exposure percentile, whereas negative values indicate relative underperformance.

2.2 Descriptive Statistics and Initial Insights

Exploratory analysis revealed persistent spatial disparities in English proficiency across Europe. Northern European countries consistently occupied the highest positions in EF EPI rankings, while several Southern and Eastern European countries displayed lower proficiency levels.

The cleaned Eurostat dataset retained 365 country-year observations across 34 countries between 2012 and 2024, whereas the cleaned EF EPI dataset retained 238 observations across 36 countries between 2017 and 2024. After temporal lag alignment and dataset merging, only 165 complete country-year observations remained available for direct comparison across both datasets.

Temporal evolution appeared relatively stable throughout the observation period. Although certain countries exhibited moderate improvements or declines, the overall European hierarchy remained comparatively stable between 2017 and 2024.

Percentile standardisation revealed substantial heterogeneity in the relationship between ESL educational exposure and observed proficiency outcomes. Certain countries achieved relatively

strong proficiency despite lower-than-expected exposure indicators, whereas others underperformed relative to their educational exposure levels.

The derived variable `gap_pct` displayed both positive and negative deviations across European countries, highlighting substantial differences between educational exposure and observed English proficiency outcomes. Percentile values ranged between approximately 0.03 and 1.00 for both ESL exposure and EF proficiency indicators.

Interactive choropleth maps and temporal trend visualisations further revealed regional clustering patterns and long-term stability in proficiency disparities across Europe. Northern European countries generally displayed positive proficiency deviations relative to educational exposure, whereas several Southern and Eastern European countries exhibited comparatively lower relative performance.

2.3 Identification of Missing Data, Noise, Bias, Inconsistencies, and Limitations

Several methodological limitations affect the interpretation of the results.

First, EF EPI data is based on voluntary online participation rather than probabilistic sampling. This introduces self-selection bias, since participants are more likely to be individuals already interested in testing their English proficiency. Consequently, EF EPI scores may not perfectly represent national populations.

Second, comparability across countries may be partially affected by heterogeneous participation rates, internet access, and demographic differences between respondents. Although the analysis assumes that these biases remain relatively stable across countries and years, this assumption cannot be fully verified.

Third, the observational nature of the project prevents causal inference. The visualisations and comparisons only identify associations and relative patterns between ESL exposure and proficiency outcomes.

Additional limitations arise from temporal disruptions linked to the COVID-19 pandemic, which may have affected both educational exposure and language acquisition dynamics during specific years.

From a technical perspective, substantial inconsistencies and missing observations were present across both datasets. The raw Eurostat dataset initially contained a large number of missing values and heterogeneous country identifiers, including regional entities and incomplete observations. Several preprocessing operations were therefore required, including country harmonisation, invalid identifier filtering, and removal of incomplete country-year combinations.

The temporal lag alignment strategy additionally reduced the number of complete observations available for analysis. Although the merged analytical dataset contained 365 country-year observations, only 165 observations corresponded to complete matched records across both datasets after preprocessing and lag alignment, corresponding to an effective matching rate of approximately 45%.

Finally, percentile standardisation improves cross-country comparability but also reduces sensitivity to absolute differences in English proficiency levels. Consequently, the derived `gap_pct` variable should primarily be interpreted as a relative comparative indicator rather than an absolute performance measure.

3 Conclusion

The development of this project required the integration and harmonisation of multiple heterogeneous datasets originating from different public institutions, reporting systems, and spatial scales. Across all sections of the project, substantial preprocessing operations were necessary,

including data cleaning, country and regional harmonisation, temporal alignment, variable standardisation, and the treatment of incomplete observations.

The implementation process also highlighted several practical challenges commonly encountered in data visualization projects involving heterogeneous open datasets. Differences in temporal coverage, missing observations, varying spatial granularity, and inconsistencies in reporting methodologies required methodological adaptations compared with the initial project proposal.

Because the project remains exploratory and observational, important methodological limitations persist. Several datasets contain potential sampling biases, heterogeneous coverage, or measurement inconsistencies across years and regions. Consequently, the visualizations should primarily be interpreted as descriptive and exploratory tools rather than instruments for causal inference.

Despite these limitations, the project demonstrates how interactive visualization techniques can support the exploration of complex educational and socio-economic patterns across Europe and Italy. The final dashboard enables users to compare regions, investigate temporal trends, and explore potential relationships between English proficiency indicators and broader contextual variables through an accessible visual interface.

More broadly, the project highlights both the analytical potential and the practical challenges associated with combining heterogeneous public datasets within a unified interactive data visualization framework.

The complete reference list and broader theoretical context are available in the accompanying project proposal document.